

CENTRAL CALIFORNIA
EMERGENCY MEDICAL SERVICES
A Division of the Fresno County Department of Public Health

Manual:	Emergency Medical Services Administrative Policies and Procedures	Policy Number: 510.06
Subject:	Basic Life Support (BLS) Protocols	Page: 1 of 3
	SPINAL MOTION RESTRICTION	
References:	California Administrative Code, Title 22, Division 9, Chapter 3.1	Effective: 11/15/83

I. SPINAL MOTION RESTRICTION

A. Goal:

The goal of Spinal Motion Restriction (SMR) is to prevent secondary injury and worsening morbidity in patients with unstable spinal injury, reserve full spinal precaution use for high-risk patients, and avoid complications associated with full spinal immobilization. Decrease/minimize use of backboards.

B. Indications for SMR in blunt trauma with suspected spinal injury include:

1. Acutely altered level of consciousness (GCS<15) or evidence of intoxication such that their appreciation of pain or ability to communicate is impaired in the setting of trauma.
2. Significant midline neck or back pain/tenderness in the setting of trauma.
3. Focal neurologic deficits (numbness/paresthesia or weakness) in the setting of trauma.
4. Anatomic deformity of the spine secondary to trauma.
5. Distracting circumstances/injury that impairs the patient's ability to contribute to a reliable exam in the setting of trauma (i.e. long bone fracture, significant pelvic injury, severe head injury).

If the above criteria are not met, but there is clinical concern for spinal injury, the patient should be placed in SMR.

C. Contraindications:

1. Isolated penetrating trauma.
2. Facial/oral bleeding or uncontrolled vomiting such that the airway cannot be controlled with SMR in place.
3. Uncontrolled hemorrhage in the face or neck area that cannot be managed with SMR in place.

Approved By: EMS Division Manager	Daniel J. Lynch (Signature on File at EMS Agency)	Revision:
EMS Medical Director	Miranda Lewis, MD (Signature on File at EMS Agency)	09/01/2025

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D. Procedure:

1. Spinal motion restriction does not take precedence over airway, respiratory, and cardiovascular stabilization in the critical trauma patient.
2. Assess for possibility of spinal injury by performing history and exam:
 - a. Check wrist/finger flexion (both hands), plantar/dorsiflexion (both feet).
 - b. Check gross sensation in all extremities and evaluate for paresthesia.
 - c. Palpate the midline cervical, thoracic, lumbar spine to evaluate for tenderness.
3. Apply appropriately sized cervical collar while a second provider maintains inline stabilization of the cervical spine while avoiding traction.
4. Moving the head to a neutral inline position is contraindicated if:
 - a. There is pain upon starting movement.
 - b. There is muscle spasm or resistance to movement.
 - c. The head is held rigidly to one side.
5. The remainder of the spine should be stabilized by keeping the head, neck, torso in alignment. This can be accomplished by securing the patient supine on a long back board, scoop stretcher, breakaway flat, vacuum mattress, or on the ambulance gurney. Lateral head support with blocks or towel rolls should be used when possible.
6. Allow ambulatory patients to sit on the stretcher and then lie flat (no standing take down).
7. If there is prolonged transport time and patient is alert and cooperative, the extrication device (i.e. long back board) may be removed once the patient is safely positioned on the ambulance gurney provided there are adequate personnel available to minimize unnecessary movement of the spine during removal. Do not delay transport for removal of the long back board in the critical trauma patient. If the long back board is removed, SMR must still be maintained with the patient firmly secured to the gurney with seatbelts and cervical collar in place. A moving device should be placed under the patient to facilitate movement off the EMS gurney on hospital arrival.
8. Reassess and document sensation and motor function following SMR.

E. Pediatric patients:

1. The approach to spinal motion restriction is similar for pediatric patients. The decision to apply SMR should not be made based solely on age.
2. Indications for SMR in blunt trauma include:
 - a. Complaint of neck pain, inability to move the neck, or midline tenderness in the setting of trauma.
 - b. Focal neurologic deficit (numbness/paresthesia or weakness) in the setting of trauma.
 - c. Altered mental status in the setting of trauma.

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d. Involvement in high-risk motor vehicle collision, high impact diving injury, or substantial torso injury.

3. Appropriate patients may be secured to the gurney in their car seat. If an appropriately sized collar cannot be safely fitted, the spine may be stabilized using a combination of towels, soft collar, or lateral head support.

F. Special Considerations:

1. SMR does not take precedent over airway, respiratory, and cardiovascular stabilization. Rapid extrication and transport remain top priority for the critical trauma patient.
2. SMR in penetrating trauma is associated with increased mortality. SMR should not be used in isolated penetrating trauma.
3. A poorly fitting cervical collar is worse than no collar at all. Distraction or hyperextension of the cervical spine due to a poorly fitting cervical collar may lead to worsening injury.
4. Patients with severe kyphosis, size constraints, or other conditions causing anatomic deformity of the spine (i.e. ankylosing spondylitis) may be stabilized using a combination of lateral head support with blocks or towel rolls, blankets, soft collar, vacuum splint or other devices if an appropriately sized cervical collar cannot be fitted.
5. Cervical immobilization is not necessary for patients complaining of isolated lumbar pain/tenderness who do not have a neurologic deficit, intoxication, altered mental status, or distracting injury present. Immobilization can be accomplished by securing the patient supine on a long back board, scoop stretcher, breakaway flat, vacuum mattress, or on the ambulance gurney.
6. If attempting to apply SMR to a combative patient would cause further detriment, abort the procedure and document in the ePCR. Notify ED staff on arrival regarding indications for SMR but the inability to apply due to combative patient.